

BIOSPARK: An End-to-End Generative System for Biological-Analogical Inspirations and Ideation

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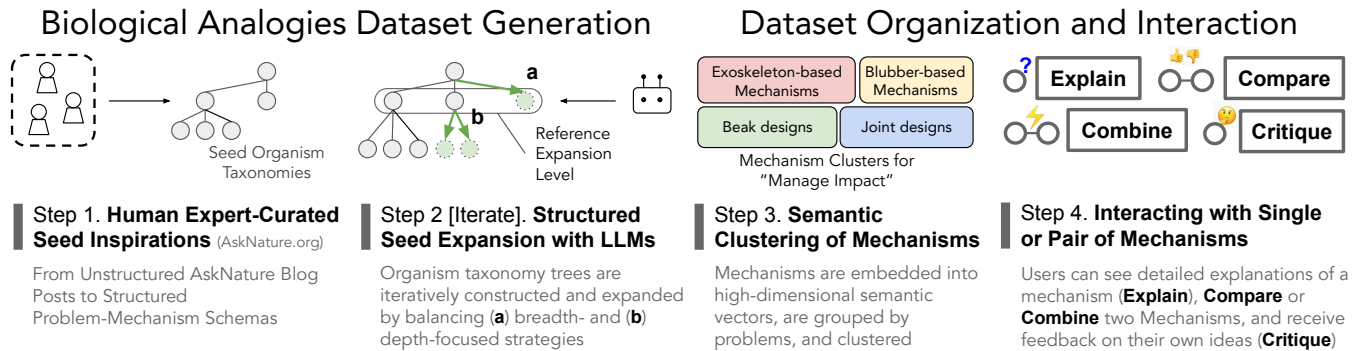


Figure 1: BIOSPARK is a system for helping designers find analogical inspirations in nature and synthesize new design insights from them. BIOSPARK consists of a Dataset Generation Pipeline and an Organization and Interaction interface. Dataset construction starts from a small set of seed mechanisms on AskNature (Step 1), expanded by iteratively constructing taxonomic hierarchies of organisms and diversifying using breadth- or depth-focused knowledge prompts (Step 2). Dataset is organized by semantic clusters (Step 3) and provided with interactive features that facilitate user learning and synthesis (Step 4).

ABSTRACT

Nature often inspires solutions for complex engineering problems, but it is challenging for designers to discover relevant analogies and synthesize from them. Here, we present an end-to-end system, BIOSPARK, that generates biological-analogical mechanisms and provides an interactive interface for comprehension and ideation. From a small seed set of expert-curated mechanisms, BIOSPARK's pipeline iteratively expands them by constructing and traversing organism taxonomies, aiming to overcome both data sparsity in expert curation and limited conceptual diversity in purely automated analogy generation. The interface helps designers recognize and understand relevant analogs to design problems using four interaction features. We conduct an exploratory study with design students to showcase how BIOSPARK facilitated analogical transfer of ideas but

was limited in conveying *active ingredients*, the core abstraction underpinning how mechanisms work. We discuss this limitation and other implications such as generative hallucination that could facilitate shifts in human exploration of new design spaces.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**; **Interaction techniques**; **HCI design and evaluation methods**; • **Information systems** → **World Wide Web**; **Users and interactive retrieval**; *Web mining*.

KEYWORDS

Analogies, Nature, Design Creativity, Ideation, Large Language Models, Diversity-enhanced Generation

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1 INTRODUCTION

Nature provides a valuable source of inspirations [3, 40, 46] for novel design solutions to challenging engineering problems [40, 43]. A central cognitive mechanism underpinning this transfer of domains is *analogical processing* [18, 20, 37], which involves discovering and recognizing deep structural similarity between source and target analogs. To this end, recent work has shown that analogs with a similar purpose (*i.e.*, ‘*what problem it solves*’), but diverse mechanisms (*i.e.*, ‘*how it solves it*’), facilitate creative new ideas [26, 30].

Yet, the full potential of bio(logical)-analogical inspirations in engineering and design has proven difficult. Many bio-inspired designs bias toward well-known organisms [6] and prior work has shown an average of three person-hours to sift through vast, fragmented information sources such as Google Scholar, Wikipedia, AskNature.org, or the open Web to find a relevant article for a design problem [44, 45]. Human-expert-curated databases such as AskNature [12], DANE [22], SAPPHIRE [9], BioTRIZ [46], or rule-based databases (*e.g.*, [10]) can streamline searches for bio-analogical mechanisms, but the effort-intensive process of manual expert curation often results in data scarcity, limiting the diversity of organisms and mechanisms available for making analogies. Recent advancements in AI, such as Large Language Models (LLMs) provide promising new avenues for scaling valuable bio-analogies. One approach is purely automated generation of analogies with LLMs, for example by iteratively prompting them to curate an ever-increasing dataset. However such zero-shot prompting approaches still resulted in limited diversity on abstract concepts [11]. To mitigate the lack of conceptual diversity in generated output, LLM prompts may be augmented with human knowledge to guide generation. Knowledge-augmented prompting, previously explored for increasing the factuality of LLM-generated answers in Q&A scenarios, provides an alternative design space to token-level manipulation [2]. The first stage of knowledge-augmented prompting is retrieval of relevant facts from a knowledge graph which then gets prepended to the input prompt to LLMs for providing factual context, enabling them to generate more accurate answers without fine-tuning or domain-specific training. In this work, we explore this design space for a different purpose of constructing a diverse biological mechanisms dataset, with a modular and structured knowledge augmentation approach.

Beyond sourcing biological analogies, it can be hard for people to understand and recognize their applicability. People’s cognitive biases towards surface similarity and working memory limitations impede the mapping process between biological and engineering systems [16, 19, 39], while the need for situating found (cross-domain) analogies within the constraints and scope of the original design problem adds complexity for designers developing their own ideas (*cf.* [21, 29]). To mitigate these challenges, we construct a scalable analogy pipeline that balances the cost of expert-curation and the limitations of naïve LLM-based generation, and develop an interactive system that aids designers in recognizing and understanding the relevance and adaptability of analogs for a chosen design problem.

2 USAGE SCENARIO

Consider an automobile designer working on a new bike rack design for sedans. The designer identifies the core challenge of designing a secure bike rack as turbulence management, caused by variable wind resistance due to changing vehicle speeds and weather. When the designer clicks on the ‘Manage Turbulence’ problem as a query (input box in Fig. 2, top left), BioSPARK displays mechanism clusters related to the problem. She quickly identifies particularly interesting and idea-provoking mechanisms such as “Adaptive shape-shifting mechanism in intertidal microalgae” and “Friction-based attachment mechanism of parasitic copepods,” two quite different mechanisms, yet commonly observed in organisms that live underwater. When exposed to turbulence cues, shape-shifting microalgae adaptively reconfigure their surface shapes to reduce resistance from strong currents. This provides her a design idea for adaptively “lying down” bike racks on the roof of the car that reduce air drag, and a “tail basement” bike rack attachment to the back of the car that joins the rotational axis of the front wheel of the bikes and adjusts during turbulence by flexibly (counter-)swinging bikes, mimicking bird tail motions. She clicks on the ‘Combine’ tab with the two mechanisms selected to experiment with design ideas. The result describes an approach that simply combines the friction-based mechanism to the shape-shifting mechanism. This gives her an additional idea for further securing the racks, by incorporating designs inspired from the adhesion mechanism of parasitic copepods, such as microridges and channels for increasing friction upon contact.

3 SYSTEM DESCRIPTION

In our paper, we introduce a system for deriving and diversifying biological mechanisms from curated blog posts on AskNature, which serve as a seed set for our database (Appendix A.1). Utilizing GPT4, we structure the unstructured blog post text into succinct problem-mechanism schemas. We then expand our dataset using a two-tier taxonomic tree approach, prompting GPT4 to explore both breadth and depth-first strategies for finding new mechanisms, resulting in a structured list of mechanism-organism pairs (Appendix A.2). Semantic clustering is applied to group mechanisms for users’ convenience, using k-Means to organize them into semantically related collections based on high-dimensional vector embeddings (Appendix A.3). To visually support the textual descriptions, we retrieve representative images using search queries composed of the organism name and mechanism description (Appendix A.4).

To assist designers in interacting with these inspirations, our system includes features for explanation, comparison, combination, and critique of mechanisms using GPT4-generated content within the interface (Appendix A.5). These features allow users to delve into individual mechanisms, directly compare mechanisms, synthesize new ideas by combining mechanisms, and critique their own design ideas, providing a rich environment for design exploration.

4 EXPLORATORY STUDY

Using the BioSPARK as a probe system, we ran an exploratory study with four participants (all male, avg. 29.0 age: SD: 7.39) to uncover design goals for further research on facilitating design inspirations with bio-analogy mechanisms. Our participants had backgrounds in : mechanicalengineering, CAD (2), visual, communication & UX

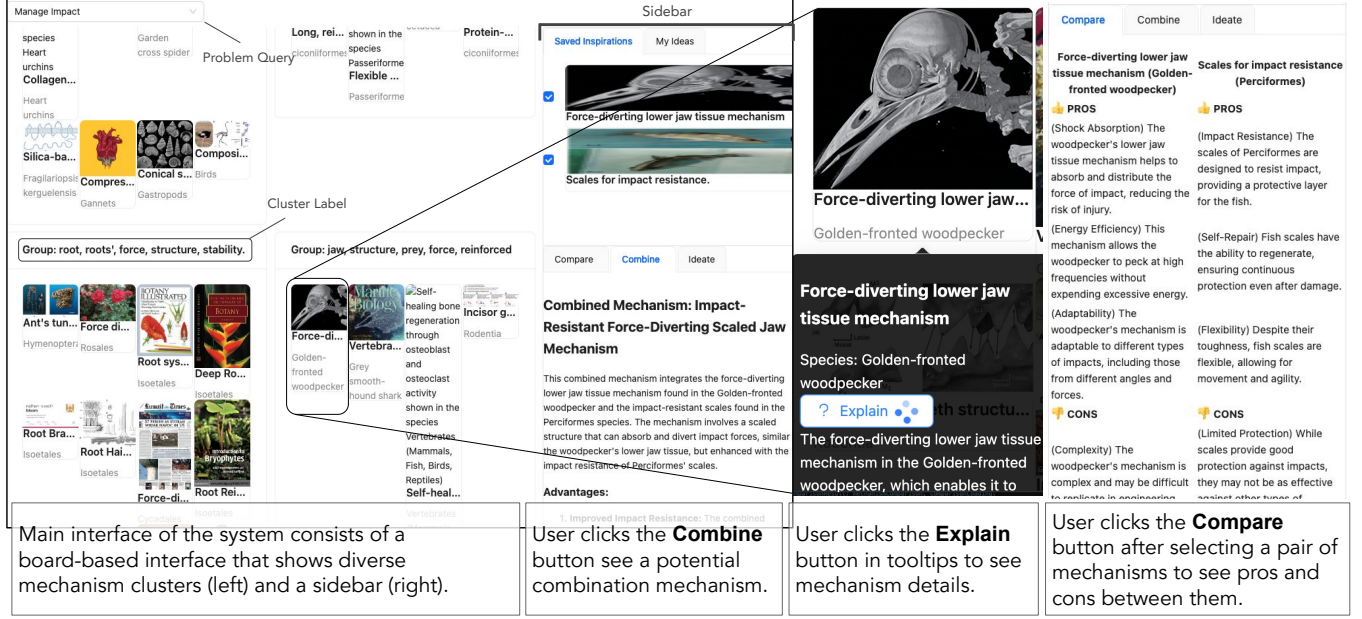


Figure 2: BioSPARK interface and a subset of available interaction affordances.

design (1), and creative coding and visual design (1). Our research goals were 1) to understand how designers might use the mechanisms to come up with creative new ideas; and 2) to understand current limitations of the mechanisms for facilitating new design insights. We anticipated that designers who are less familiar with biology may find it difficult to engage with diverse organism species and their mechanisms, and thus added a stage in the study where we generated synthetic mechanism visualizations to probe related challenges.

4.1 Procedure

The study took place virtually on Zoom. First, participants completed a short demographic survey and were given a tour of the BioSPARK interface and its features. The experimenter then introduced the main ideation task in which participants were given one of two randomly selected design briefs ('1) design a secure bike rack for sedans' or '2) design improvements for sedans on slippery road conditions') and asked to use BioSPARK to find as many inspiring mechanisms to come up with new design ideas 20 minutes. Participants shared their screen and thought aloud during the task, and were encouraged to use all of the system features to come up with at least two new design ideas for the brief. Participants self-recorded their ideation outcomes and inspiring mechanisms found on BioSPARK using a shared document. After the design task, the interviewer presented a series of synthetic images representing mechanism inspirations each participant found and asked for their critique about their helpfulness in facilitating generation of new design ideas, and the ease and challenges with understanding the mechanisms. The study lasted around one hour.

4.2 Synthetic Mechanism Visualization

We employed two strategies for synthetic mechanism visualization in biological organisms. Firstly, we used direct prompting to generate diverse images from a text-to-image AI model, Dall-E3 [4], based on the mechanism description and organism name (details

in Appendix B.1). Sample visualizations are provided for 'Body shape streamlining in coleoptera' (Fig. 4, left). Secondly, to address the need for more focused visual differentiation, we developed an aspect- and vantage-point-focused generation approach (we describe our motivation in Appendix B.2). Here, we first identify up to four unique visual aspects of the mechanism using GPT4 and then generate corresponding images with Dall-E3, emphasizing one aspect at a time (details in Appendix B.2 with an example visualization in Fig. 4, right). This method led to clearer mechanism visualizations that highlighted specific aspects with corresponding vantage points.

4.3 Findings

4.3.1 Value 1. BioSPARK mechanisms facilitated analogical transfer. All participants found several BioSPARK mechanisms to be inspiring sources of new design ideas. P3 was inspired by tentacle coiling of octopi and flexible lizard tails to ideate a design concept of modular bike rack components that could gracefully expand and contract based on air flows and turbulence levels. P2 was inspired by scale and fur arrangements for squamata and rodentia, respectively, to ideate new groove patterns on tires that could dynamically 'stand up' when the vehicle is turning to create more downforce and therefore more traction on slippery roads. The participant also noted that this idea could be applied to vehicle vent grilles and spoilers for similar effects.

4.3.2 Value 2. Counter-intuitive mechanisms for novel transfer. Participants were surprised by counter-intuitive mechanism inspirations and were also intrigued by their potential for facilitating novel transfer of ideas. For example, P1 mentioned that "(bird) feathers are thought-provoking to me from the materials point of view because it's not a sturdy or solid material which is the first thing I thought about for this design problem." P2 also noted how "fish shapes or fluid dynamics are interesting... they definitely spark new associations and ideas because 'fluid turbulence' (for fish) and 'air turbulence'

(for automobiles) are related.” Participants engaged with counter-intuitive mechanisms as ingredients for novel transfer despite their varying levels of expertise in automobile design. For example, P1, with zero experience in automobile or mechanical engineering design albeit several years of experience in UX and interaction design, found many BIOSPARK mechanisms counter-intuitive and thought-provoking (e.g., “I would have never thought to look at slime mechanisms for bike rack designs, but they make me think about the attachment aspects.”). At the same time, P2, who was experienced in PhD-level mechanical engineering design projects, analyzed the design brief with first principles of physics before engaging with BIOSPARK mechanisms (“In order to prevent vehicles from slipping on bad road conditions, we need to increase downforce or manipulate the interfacial friction between the solid and the liquid.”). The analysis provided him a frame for what mechanism qualities he would look for (i.e., mechanisms related to increasing downforce or interfacial friction), which guided his filtering efforts. Nonetheless, P2 found multiple species and their mechanisms to inspire novel design ideas. For example, upon learning how Anuras modulate their skin texture to manage water turbulence, P2 was triggered to think about new designs for tire grooves that could modulate friction upon contact with liquids. In addition, learning about how Xenarthras’ body shape allowed efficient burrowing made him think about adjustable tire widths that could improve the robustness of cornering.

4.3.3 Challenge 1. BIOSPARK mechanisms need to highlight active ingredients. Participants noted that visual representations of mechanisms were important to understand and transfer them, but BIOSPARK’s were inadequate because they lacked focus on **active ingredients**, the core abstraction underpinning how each mechanism actually works. While existing BIOSPARK mechanisms commonly included whole body images of organisms, participants commented that showing active ingredients such as “feathers and feather constellation patterns rather than the whole body image of birds” (P1); “skin texture modulation and surface patterns involving fur” (P2); and “curvature of the tail and claws in geckos” (P3) would be better for transferring a concept.

4.3.4 Challenge 2. Visualization for effective mechanism comprehension & transfer. To this end, participants commented that **incorporating multiple spatial and time vantage points in the mechanism visualization** without highlighting irrelevant aspects would be helpful. P4 wanted images to “zoom into the microstructures (because) at higher scales, when you look at the whole body, you cannot get the sense of the actual mechanisms”. P3 similarly commented “The first image is pretty cool, zooming into the surface, the layout, the skin patterns... which gives me ideas about mathematical and surface patterning (that could be applied to cars)”. Along with different scales of zooming, variations of multiple vantage points could help to give a sense of volume and facilitate ideation: “The middle image has more abstract shapes shown in different vantage points, which seem helpful in earlier stages of the design process when designers need to consider different shapes for streamlined bodies” (P2). Relatedly, P3 commented that “2.5D/3D images supplementing 2D images is a very good idea, it’s helpful for mapping the mechanism ideas onto the actual 3D car models and incorporating them”.

4.3.5 Challenge 3. Envisioning how mechanisms could be transferred to target domains. Participants also wanted additional support for **envisioning mechanism inspirations transferred into target design domains**. P2 mentioned that transferring mechanism concepts related to aerodynamic efficiency or turbulence management in automobile designs may require “getting certain motives out of the images... picking out a few lines from them and applying them (to car designs)” (P2). P3 commented how rendering mechanism images in 3D could help close the visualization gap for designers: “it’s helpful for mapping the design onto the actual 3D car models... 3D objects would help with that incorporation and design execution”. Participants also commented that generating images for incorporation of mechanism ideas into focused areas of target domains may facilitate engagement with more distant ideas that have potential for high impact when iterated on. P2 said: “I think more targeted treatment, focused in scope would be good. Maybe we (users) can apply scales to only parts of the vehicle, such as wheel rims or front grilles or spoilers”. Furthermore, P1 described how support for further exploration of a mechanism idea could be useful: “‘Bike rack’ and ‘slime’ are somewhat contradictory but it (slime mechanism) makes me think about the attachment aspects of the design... maybe new ideas around loading and unloading of bikes that have dynamically adjusting surface friction... I’m going to click ‘explain’ on slime... (after the detailed explanation loads) I wish I could know more about the lubrication mechanism aspect of slimes”.

5 EVALUATION OF THE BIOSPARK MECHANISM DATASET PIPELINE EFFICACY

Complementing the exploratory study results, we also evaluated the efficacy of different components that consisted the BIOSPARK dataset pipeline, such as the accuracy of LLM-based taxonomy construction (Appendix C) and the increase in the diversity of represented organisms in the dataset (Appendix D), which we detail in the Appendix.

6 DISCUSSION AND FUTURE WORK

6.1 Identifying active ingredients related to BIOSPARK mechanisms

The findings of our exploratory study suggests further research directions around the challenge of identifying and specifying active ingredients of BIOSPARK mechanisms for effective understanding by designers and for supporting abstraction and transfer of mechanistic concepts between domains. One research direction may be leveraging BIOSPARK mechanism dataset pipeline as a way to expand seed queries whose output can be input to a different process for finding more detailed descriptions of active ingredients and their provenance on the web. For example, while a BIOSPARK mechanism “The feathers of galliformes are arranged in a way that optimizes airflow and minimizes turbulence” pointed to an interesting design space of feather constellation for P1, it did not convey how feathers optimize airflows in sufficient details nor how could the participant abstract and adapt that idea to designing new bike racks for automobiles. In contrast, by framing this seed mechanism as a question “How do galliformes manage air turbulence?” and searching on Google returned a research article about related but different species, Jackdaw. The article described how its wings

form bendy articulated tips during flight that create air vortex to improve aerodynamics. While Jackdaw belongs to the bird order of Passeriformes, not Galliformes, both are part of the Aves class based on their physiological similarities (e.g., feathers, toothless beaked jaws, a strong yet lightweight skeleton). The existence of such similarities suggests BioSPARK-generated mechanisms can be useful for navigation to approximate information sources that contain the level of details in descriptions that include active ingredients (e.g., 'bendy articulated tips for air vortex creation') to help designers more effectively understand and adapt the idea to their own domains.

6.2 Hallucinations, Plausible mechanisms, and Shifting human exploration of design spaces

In our exploratory study results, the primary type of value from BioSPARK mechanism inspirations was its facilitation of analogical transfer. Participants engaged with BioSPARK to oft-find counter-intuitive mechanisms that pointed them to new design spaces and challenge preconceived assumptions about problems. Though in the present study results we did not evaluate the quality of participants' ideation outcomes (which we leave to future work), our qualitative observations of the ideation process suggested that machine-generated BioSPARK mechanisms – some of which may not be factual – contributed to expanding the bounds of human problem-solving creativity in a promising way, helping them overcome potential biases that may have constrained their search space of solution ideas.

Scaling this observation to a population level, it also relates to how machine-generated inspirations could serve as a resource for helping humans break free from conventional wisdom that sometimes prevent adoption and ideation of ideas that significantly deviate from expectation. In this regard, a recent study by Brinkmann et al. [5] noted studies on Go players that showed an unexpected shift in Go-playing culture that led to a significant upward shift in players' decision-making quality since the introduction of AlphaGo. The authors found evidence that this shift was a product of not only adoption of machine-generated Go moves that deviated from conventional wisdom, but also how human Go players started developing moves that were qualitatively different from both machine-introduced and previously human-introduced moves.

For the purpose of facilitating shifts in people's thinking in design spaces, therefore, the measures of accuracy or factuality may not be a strictly necessary condition for predicting the inspirational value of mechanisms. While their importance is clear in the feasibility and in making decisions around allocation of resources to executing particular ideas conceived from mechanism inspirations, there may also exist clear benefits from leveraging inspirations whose factuality may be unknown or those that are factually incorrect yet sufficiently evocative to human users.

In future studies, we plan to more deeply study this relation between factuality and evocativeness of machine-generated mechanisms for the purpose of augmenting human design inspiration. Here, one thread of research may explore methods for improving LLM-based content generation, such as retrieval-augmented generation (RAG; [32] showed that scholars could produce significantly higher quality with better supporting evidence from prior literature

when generating review outlines using a system with RAG capabilities), factuality-checking via automated methods (cf. [24]), or through discovering proxy experts from networks (e.g., [25, 28, 31]) and sending fact-checking help requests to them (e.g., [8, 41, 49]). We have begun to identify different types of hallucinations such as 1) attributing a wrong species to a mechanism, 2) hallucinating a plausible yet scientifically not-yet-proven mechanism, and 3) hallucinations involving both types leading to a more severe hallucination with lower factuality. We note there are important open questions related to the effectiveness of analogical transfer around how factuality relates to the plausibility of machine-generated mechanisms for receivers who may be exploring domains less familiar to them.

6.3 Closing the 'Gulf of Geometric Transfer' with combinatoric visual mechanism generation

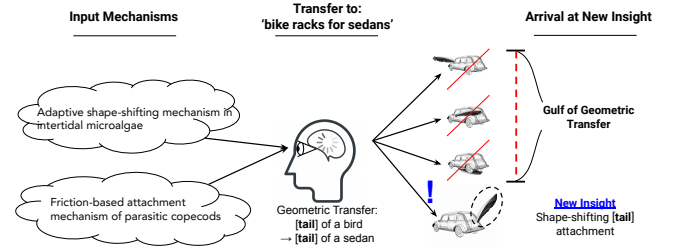


Figure 3: When the designer arrives at the new insight of designing a tail-like, shape-shifting bike rack attachment, she is combining two source mechanisms and transferring the new idea into a new geometry in the target domain ('at the tail of a sedan') that could suit the combined mechanism. The successful generation of this combination comes after a series of failed attempts to generate geometries, which we call the Gulf of Geometric Transfer.

The designer in the usage scenario arrives at a successful transfer of mechanisms naturally after several unsuccessful iterations of generation. A key process in transferring a new idea into the target domain involves visualizing a candidate geometry, such as in this case 'a bike rack attachment to the tail of a sedan'. The process of identifying an effective geometry for transfer like 'at the tail of a sedan' can resemble a combinatoric search, where the designer visualizes the location of integration for the new idea in the target object, from a possible source object and its geometry, such as 'the tail of a bird'. The process of evaluating an effective geometry for transfer may further involve analogical processing, for instance, by comparing the integrated geometry to familiar objects and judging the aesthetic and functional value of such a geometry.

We hypothesize that one challenge of this transfer process is what we term 'Gulf of Geometric Transfer' (see Fig. 3, right; the concepts related to this namesake in design and usability studies are the Gulf of Execution and the Gulf of Evaluation [35]), which refers to the failed iterations that one has to endure before obtaining a successful geometric insight that supports the new idea. To bridge this gulf, we envision a future work area that aids designers in visualizing what successful geometries that facilitate transfer of ideas might look like, and also helps them eliminate any blind spots

in the process that could hinder new insights. A fruitful starting point to this end may involve generating visual representations of geometries that are related to the new abstract idea from various key orientations, such as {up, down, front, rear, left, right, internal, external}. Enumerating these geometries visually could assist designers in effectively reviewing possible geometries and facilitating their imagination for alternative geometries that may be useful for further consideration. The combinatoric visualization of transfer could also be complemented by including other types of knowledge about the target object and constraints, such as known consumer preferences and aesthetic appeal, that would help with filtering out predictably ineffective variants during the generation process.

6.4 Supporting Generative ‘Frame Slips’

The geometric transfer discussed in the previous section can be understood as one form of a more generalized phenomenon in analogical transfer, where a conceptual frame ‘slips’ into a different one, thereby generating novel insight. For example, Niels Bohr transferred the concept of planets orbiting the sun to the domain of atomic structure to propose a new model in which electrons orbit the nucleus in a similar manner [16]. This new structure of the atom significantly differed from the dominant ‘plum-pudding’ model of the time, leading to a *re-structuring* of the concept [17]. Here, *re-structuring* is related to conceptual frames that successfully ‘slip’ into a new, analogous problem domain, causing structural changes in the conceptual framework of the receiving domain or vice versa. In the case studies, the concept of the ‘tail of a car’ slipped into geometrically and syntactically similar ‘tail of a bird’, engendering a new insight. Similarly while studying how crowds generate design concepts for children’s chairs, Yu et al. [48] found that the concept of a playground ‘slide’ slipped into another semantically related playground concept ‘swing set’, which then gave a novel design idea of swing chairs.

Understanding the process of ‘Frame Slipping’ is therefore crucial for improving analogical transfer. Existing examples of ‘frame slips’ imply that such transitions can occur through different relational paths, for example, syntactic, semantic, functional, abstracted commonality, geometric, or other means. While geometric transfer may impose a more concrete form of transformation onto the transfer process, other means of transfer may be less understood, possibly involving more complex multimodal processes that combine multiple relational paths at once. Nonetheless, significant value of an inspiration often results from enabling frame slips to new parts of the design space, opening up not only a single point of an idea, but an entire space of ideas for future exploration.

7 CONCLUSION

In this work we present BIOSPARK, an end-to-end system for generating a biological-analogical mechanisms dataset and an interactive interface that facilitates learning new biological mechanisms for design challenges and synthesizing new solution ideas inspired by analogical mechanisms. We imagine a future in which engineers and designers could find inspirations based on deep analogical similarity between mechanisms found in nature to problems common to engineering and design challenges to drive innovation across

fields. Future work remains in this area to provide improvements on the pipeline for generating analogical inspirations from nature, and in supporting users’ recognition and synthesis from them for materializing downstream innovations.

A SYSTEM IMPLEMENTATION DETAILS

A.1 Structuring AskNature blog posts’ text into a seed set of problem-mechanism schemas

To source a set of diverse, high-quality biological mechanisms for a given problem, BIOSPARK starts from a seed set of expert-curated biological mechanisms on AskNature (Fig. 1, Step 1). AskNature.org provides a curated list of organisms with detailed descriptions of their unique strategies to functional problems (e.g., ‘Manage Impact’, ‘Modify Speed’). The organisms and strategies can be grouped by function and viewed as a list. To curate a seed set of high-quality mechanisms, we first choose a functional problem p predicted to be highly relevant to automobile designers, excluding irrelevant functions such as ‘Adapt Behaviors’, ‘Adapt Genotype’, ‘Coevolve’, ‘Maintain Community’. We access the sub-list of organisms $o \in O$ and strategies posted to p on AskNature’s group-by-function page by parsing the HTML code using the BeautifulSoup package on Python. We then access the blog post for each organism-strategy page using the parsed URL and parse the returned HTML page to get the title, description, and references (if available).

At this stage, the returned unstructured text is yet to contain a succinct mechanism description. Furthermore, we found that some blog posts do not contain any body text despite having a title and are accessible via the URL. Some of these missing blog posts indicated that they are in maintenance and/or planned to be updated. To structure the raw blog post text $\text{AskNature}_{(o,p)}$, we prompt GPT4 [36] to succinctly describe (i.e., using 12 words or less) the core mechanism (i.e., excluding the qualities or effects, and focusing on mechanisms with engineering design implications), given (o, p) (if blog post text is missing) or $(o, p, \text{AskNature}_{(o,p)})$. The returned mechanism description m along with the function description makes up the problem-mechanism schema for each organism: $\{o \in O | (p, m, o)\}$.

A.2 Iteratively expanding mechanisms dataset by traversing constructed taxonomic trees

Using each schema as a seed, we iteratively prompt GPT4 to find relevant mechanisms for the given mechanism and problem, using an even mixture of breadth- and depth-focused expansion strategies (Fig. 1, Step 2). To enable structured diversification of organisms and their mechanisms beyond prior work that relied on token-level manipulation or naively prompting LLMs, we guide LLMs *how* and *where* to expand by leveraging organism taxonomic hierarchies. At each iteration of expansion (Fig. 1, Step 2), we aggregate the organisms represented in found mechanisms up to that point, and construct a taxonomic tree featuring seven levels of hierarchy on Tree of Life: {domain, kingdom, phylum, class, order, family, genus, species}, where domain representing the highest level and species representing the lowest level on the hierarchy.

Given this tree, we aim to identify sparsely populated branches for expansion. We cut the tree at a given reference expansion level

(e.g., class), and sort the taxonomic ranks (nodes) on that level by the number of its immediate children nodes¹, in an increasing order. For performance, we batch 10 prompts to send to GPT4 for expansion. For half of the prompts, we instruct **breadth-first expansion** which asks GPT4 to first identify *sibling* nodes at the given reference taxon level and existing nodes (up to 50 most populated nodes). For example, the prompt asks “come up with a few biological classes not in {...names of excluded classes...}”. The breadth-first expansion prompt then instructs GPT4 to repeat the following: 1) Choose one taxon from the list it came up with; 2) Succinctly describe (i.e., using 14 words or less) new mechanisms m related to a problem p . For the rest of the prompts, we instruct **depth-first expansion** which asks GPT4 to first identify a new *children* node at the given reference taxon level and existing children nodes (up to 50 randomly sampled children). For example, the prompt asks “come up with a few biological families in order araneae that are not any of {araneidae, ...}”. The depth-first expansion prompt then instructs GPT4 to repeat a similar procedure as breadth-first expansion. In the prototype system, we run 10 batches for expansion to construct dataset of mechanisms for each problem.

The returned list of mechanisms and organisms text are then fed into the second GPT4 prompt for structuring them into a list of {mechanism, organism} dictionaries. Finally, using each organism name, we prompt GPT3.5-turbo to retrieve the seven-level taxonomic hierarchy, based on our model evaluation result showing its high accuracy (Appendix C).

A.3 Semantic Clustering of Mechanisms

The mechanism descriptions are clustered in semantically related groups for designers’ efficient viewing (Fig. 1, Step 3). We first embed the mechanism descriptions for a given problem into high-dimensional vectors using the text-embedding-ada-002 end-point of the OpenAI API. We group the embeddings into $k = 20$ clusters using k-Means clustering. Each group label is generated by chunking the mechanism descriptions in each cluster into lower-cased words, excluding stopwords. We count the frequency of each word and take the top 5 most frequent words as a description of each cluster.

A.4 Representative Mechanism Image Curation

To aid designers’ visual understanding of and pique curiosity for biological-analogical mechanisms, we retrieve representative images for corresponding textual mechanism descriptions. We use Google Custom Search² with queries as “[organism name]: [mechanism description]” and the file type set to images and the safe search mode enabled. We choose the first place result of Custom Search as the visual representation of each mechanism.

A.5 Interacting with Mechanism Inspirations: Explain, Compare, Combine, and Critique

To facilitate designers’ understanding and synthesis of mechanism inspirations, we develop several interaction features available on the interface (Fig. 2). The **Explain** button is located in tooltips that

pop up when the user places the mouse over on a mechanism card in the board UI (Fig. 2, first panel). When the user clicks on the button, BioSPARK sends a prompt to GPT4 requesting elaboration of the interacted mechanism and the organism in the context of the chosen engineering design problem. The **Compare** tab is located in the control bar of the sidebar of the interface. To use this, users need to first click on (at least) two mechanism cards from the left, saving them to the ‘saved inspirations’ panel at the top of the sidebar. There, users can check any two of the saved mechanisms they wish to compare. BioSPARK sends a prompt to GPT4 when the user clicks on the tab, requesting comparison of pros and cons between the two mechanisms in the context of the chosen engineering problem. The result is formatted into a markdown table, with each mechanism as the header followed by pros and cons rows detailing each point. The **Combine** tab is also located in the control bar of the sidebar in the interface. Similarly with Compare, users can check two saved mechanisms they wish to see combined. BioSPARK sends a prompt to GPT4 then requesting elaboration of a mechanism that combines the two selected mechanisms, and explain its potential advantages in the context of the chosen engineering problem. The result is also formatted into a markdown page using section title and headers for demarcating the content. Finally, the **Critique** button is located inside the Ideate tab. Upon clicking the Ideate tab, users can type in their own idea in the rich text editor in the opened tab, and optionally click on the button below to receive constructive feedback on it. BioSPARK sends a prompt to GPT4 with the content of the text editor describing the idea, and requests additional revision that may improve the quality, such as anticipated failure modes and potential improvements.

A.6 Extending BioSPARK to support any problem queries by incorporating rich problem-mechanism relations

One of BioSPARK’s limitations is its fixed problem queries. Though the five pre-generated problem queries provide a useful entry to mechanism organisms that may be applicable to a diverse set of design challenges, it comes at the cost of an inability to query biological mechanisms for *any* engineering design problems described in natural language text. As designers and engineers progress in interacting with the system, they may naturally come up with follow-up queries that may differ from the source queries, that could emphasize important constraints around the design problem, or specify low-level details newly understood to be important to consider. Adaptation to such evolving user query intent requires further personalization and scaffolding in the workflow. In future work, mixed-initiative workflows may leverage user interaction traces as input to LLM operations (cf. [32, 33]) to augment query input and automatically search data to retrieve analogical results.

In order to enable search by free-text problem queries, the underlying data model needs to be extended to contain multiple problem-mechanism relations beyond the single problem present in the schema $\{\forall i|(p_i, m_i, o_i)\}$, and into an enriched dataset with mappings between problems $p_1, p_2, \dots, p_n \in P$ and an applicable mechanism m_k , as commonly the case in engineering (e.g., ‘spider silk’ can be used for multiple engineering challenges such as replacing steel bars in concrete or wound suture and prosthesis [23]).

¹alternatively, the entire size of the subtree, rather than immediate children, could be used for sorting

²<https://developers.google.com/custom-search/v1/overview>

One way to expand the rich problem-mechanism relations in a scalable manner is to prompt LLMs to come up possible engineering design problems that a given biological mechanism could be applied to. Here, naïve prompting may suffer from conceptual redundancy, analogous to the challenge of curating diverse mechanisms, that limits the diversity in mechanism-to-problem mappings. Another approach may be to intelligently use the existing dataset $\{\forall i|(p_i, m_i, o_i)\}$ to identify similar mechanisms m_i in (p_i, m_i, o_i) and m_j in (p_j, m_j, o_j) that can be mapped onto disparate problems: $m_i \sim m_j \rightarrow p_i, p_j$. This approach however assumes the presence of many correlated such mechanisms with disparate problem pairs in the dataset, which need empirical examination for support. Once the enriched dataset $\{\forall i|m_i \rightarrow S(m_i)\}$ (Here, $S(m_i)$ denotes the set of engineering design problems that m_i is applicable to) is made available, one simple approach for allowing querying on any problem text is to construct a similarity search index (e.g., the HNSW index of FAISS [27]) using a chosen text embedding approach.

B SYNTHETIC MECHANISM VISUALIZATION

We explored two complementary approaches for visualization.

B.1 Direct Prompting Approach

In the direct prompting approach for synthetic mechanism visualization, we prompt Dall-E3 using a combination of the description of the mechanism and the species of the organism using the template in Fig. 5. We set the image resolution for generation to 1024x1024 in size and standard quality resolution. We set the ‘style’ parameter to vivid, which guides the model to “lean towards generating hyper-real and dramatic images”³. We repeatedly sample 3 images to incorporate visual diversity among the images.

B.2 Aspect- and Vantage-Point-Focused Prompting Approach

We noticed that the simple repeated sampling approach incorporated variations among the generated images focused on visual aesthetics. However, we also noticed that the variations did not sufficiently incorporate diversification based on different aspects involved with each mechanism. For example, though differing in the specific color palette or orientation used for depicting the organism (Fig. 4, left), the images commonly featured an anatomical drawing of the whole body, using a top-down perspective, similar to what might be expected in displays in natural science museums. We also noted that generated images rarely focused on a single aspect or view point of the mechanism, and instead frequently incorporated many confusing visual details such as organism anatomy into one image, potentially misleading viewers’ attention to less important aspects of the organism and the mechanism. Motivated by this, in the second approach we aimed to control for visual differentiation by focusing generation along one relevant aspect of the mechanism at a time. To this end, we developed a two-step generation pipeline: In the first stage, we prompt GPT4 to generate up to four unique aspects about a mechanism using an aspect- and vantage-point-elaboration prompt with both the mechanism description and the

organism name. The prompt asks GPT4 to describe how the mechanistic aspect described could be best visually displayed using a particular vantage point for the species and to format the aspect descriptions into a list of JSON objects. In the second stage, we iteratively prompt Dall-E3 using each JSON object and the same text-to-image generation prompt (Appendix B.1) as before to generate mechanism visualizations along a single salient aspect. Pilot experimentation of the outcome of this pipeline showed a clearer emphasis on individual mechanism aspects and the use of a fixed vantage point for the corresponding aspect that might visualize the aspect effectively.

C BIOSPARK DATASET PIPELINE EVALUATION: ACCURACY OF LLM-BASED TAXONOMY CONSTRUCTION

The main process in our diversification strategy is iterative construction of taxonomic trees at each stage of expansion with a set of problem-mechanism schemas and corresponding organisms $\{o \in O|(p, m, o)\}$ curated (in case of AskNature seeds) or generated up to that point. To construct the trees, the taxonomic hierarchy of each organism needs to be known. Here, we restrict our tree construction to seven levels of depth, ranging from the highest to lowest levels: domain, kingdom, phylum, class, order, family, genus, species. These levels provide considerable branch-switching opportunities for diversification, through significant changes in the number of members between levels and within each level of the hierarchy. For example, while the highest level domain consists of three members, Bacteria, Archaea, and Eukarya, there are estimated 8.7M species in the world [42]. The next level on the hierarchy, Genus, has an estimated number of 310K members [38], while the number in the subsequent level, families, is estimated at 8K [34] in 2011. The number of known species for each node on the hierarchy also changes considerably, further contributing to the diversification opportunities. For example while most non-avian reptile genera have only 1 species each, insect genera such as *Lasioglossum* and *Andrena* have over 1,000 species each, while the flowering plant genus, *Astragalus*, contains over 3,000 known species [47].

Our initial exploration of suitable approaches to retrieve organism taxonomies involved using available resources such as the Global Biodiversity Information Facility API⁴, Catalogue of Life [7], or the Encyclopedia of Life [15], where canonical species names were retrieved from the Darwin Core List of Terms⁵ for corresponding organisms in problem-mechanism schemas. However, the limited coverage, data consistency, and API availability of these tools prevented their adoption. On the other hand, Wikipedia provides scientific classification for some of the organism articles (for example in the Pomelo article⁶, taxonomic names for Kingdom, Clade, Order, Family, Genus, and Species are available in the ‘biota’ information box that appears on the right-hand side of the page). However, this data was not readily available for scalable generation.

⁴<https://www.gbif.org/developer/species>

⁵https://dwc.tdwg.org/list/#dwc_Organism

⁶<https://en.wikipedia.org/wiki/Pomelo>

³<https://platform.openai.com/docs/api-reference/images/create>



Figure 4: Synthetic mechanism visualizations using two approaches. (Left) Images generated via direct prompting on Dall-E3 for ‘Body shape streamlining found in coleoptera’. (Right) Images generated via aspect-focused prompting for ‘Body shape streamlining found in Dermoptera’. The three aspects identified by GPT4 for this mechanism were (from the leftmost image) ‘Aerodynamic Profile View’, ‘Top-Down Aerodynamics View’, ‘Microscopic Surface Texture View’, each highlighting different aspects of the mechanism (see Appendix B.2 for the full generated content).

{Mchanism Description: Body shape streamlining} found in {Species: coleoptera}

Figure 5: The prompt used in the direct prompting approach for synthetic mechanism visualization.

```
[System Message]
Identify up to three different vantage points that illuminate the mechanism described.
The vantage points for the mechanism should consider the problem context of designing a secure bike rack for sedans,
↳ where management of turbulence is critical.
The vantage points should differ in terms of the scale and the angle of focus, such as micro versus macro or portrait,
↳ side, or top views and so on.
For each vantage point, describe in detail how the mechanism aspect according to the specific vantage point can be
↳ visually drawn.
Output your response in a list of JSON dictionaries with "vantage_point" as keys and their corresponding values.
For example, [{ "vantage_point": "...", "desc": "..."}, {"vantage_point": "...", "desc": "..."}, ...].
Do not output any text other than the JSON.

[User Message]
{Body shape streamlining} found in the species {galliformes}

[Example Generated Content for the mechanism 'Body shape streamlining found in Dermoptera' (in section "Synthetic
↳ Mechanism Visualization")]
[[{
  "vantage_point": "Aerodynamic Profile View",
  "desc": "From a side view, the streamlined body shape of the gliding primate or rodent can be depicted with smooth,
↳ elongated contours that taper towards the rear. This profile would highlight the gradual reduction in cross-sectional
↳ area from the head to the tail, illustrating how the body shape is optimized for reducing drag. The limbs would be
↳ shown close to the body or extended in a way that aligns with the airflow, further reducing turbulence."
},
{
  "vantage_point": "Top-Down Aerodynamics View",
  "desc": "From a top-down perspective, the organism's body can be visualized as an elongated oval with a slight taper
↳ towards the tail end, indicating a teardrop shape that is known for its aerodynamic efficiency. The limbs, if
↳ extended, would be seen as narrow lines that either align with the body or are positioned to maintain the streamlined
↳ form. This view would emphasize the symmetry and smoothness of the body, which are critical for minimizing air
↳ resistance and managing turbulence."
},
{
  "vantage_point": "Microscopic Surface Texture View",
  "desc": "At a microscopic scale, the surface texture of the organism's skin or fur can be illustrated to show how it
↳ may contribute to reducing turbulence. The visual would depict any specialized structures, such as scales or fur
↳ patterns, that are designed to disrupt the laminar flow just enough to reduce the formation of larger, more chaotic
↳ turbulent vortices. This close-up view would focus on the intricacies of the surface that interact with the air at a
↳ very small scale, which collectively contribute to the overall aerodynamic performance."
}]
]]
```

Figure 6: The prompt used in the aspect- and vantage-point-focused prompting approach for synthetic mechanism visualization and a sample generated output. The content in curly brackets ({...}) in the user message portion were replaced with user-specific content.

C.1 Procedure

LLMs may provide an alternative solution to the limitations of existing approaches for retrieving the taxonomic hierarchy for a given organism name. To test this idea, we curated 90 gold taxonomies using Wikipedia that have complete information in the 'biota' scientific classification info box (the complete list of 90 organism names can be found in Appendix C.5). For each organism, we prompted LLMs with each organism name zero-shot using the chat completions API endpoint⁷ using each model key. The prompt used for taxonomy generation for LLMs can be found in Fig. 7. Once the hierarchy data is generated, we lower-cased the rank names for consistency.

C.2 GPT4's Accuracy

We find that GPT4's zero-shot taxonomy generation accuracy to range between 94.4% and 100% (Table 1). The lowest accuracy was observed in the family taxonomy, followed by order (96.7%) and genus (98.9%).

C.3 Error analysis

We find that some error cases in taxonomy generation could be attributed to recent changes in classification in the literature. For example, both GPT4 and GPT3.5-turbo models classified naked mole-rats as then literature-accepted 'Bathyergidae' for their family, same as other African mole-rats. However, more recently naked mole-rats were placed in a separate family, Heterocephalidae [1].

Among the error cases overlapping between the two models, we found cases that either the GPT3.5-turbo or the GPT4 model wins over the other (e.g., for 'hummingbird', GPT3.5-turbo generated 'archilochus' as its genus whereas GPT4 generated 'various'; for 'boxer crab', GPT3.5-turbo generated 'hymenoptera' which is an order of insects, whereas GPT4 generated 'decapoda', which is the correct order). In other cases, both models outputted similarly incorrect answers, for example for 'sea snail', GPT3.5-turbo generated 'neogastropoda' whereas GPT4 generated 'archaeogastropoda' (the Wikipedia gold answer was 'lepetellida').

C.4 System Optimization: GPT3.5-turbo's Accuracy

We find that GPT3.5-turbo has comparable accuracy levels with GPT4 in zero-shot taxonomy generation. The highest misalignment occurred in genus, with a 6.67% error rate (equivalent to 6 out of 90). Appendix C.3 provides a further qualitative error analysis of models' comparative performance. Based on these results, we opted for the more efficient GPT3.5-turbo model in our pipeline. We leave further exploration of the capabilities of smaller, fine-tuned base LLMs, with implications for LLM cascade⁸, to future work.

⁷<https://api.openai.com/v1/chat/completions>

⁸LLM cascade refers to a system design approach that adaptively chooses optimal LLM APIs for a given query. Smaller, task-specific LLMs are regarded as optimal when they exhibit higher or similar levels of performance compared to models that are orders of magnitude larger [13], with all else equal.

C.5 Complete List of Organisms Used for Taxonomy Generation

{'spider monkey', 'prairie dog', 'garden tiger moth', 'african sacred ibis', 'argiope argentata', 'ostrich', 'groundhog', 'danio rerio', 'gannet', 'deer', 'cattle', 'glyptodon', 'alligator snapping turtle', 'leopard', 'arctic ground squirrel', 'cormorants and shags', 'bears', 'squirrels', 'herons', 'european badger', 'golden silk orb-weaver', 'aardvark', 'seahorses', 'banner-tailed kangaroo rat', 'hyenas', 'pink fairy armadillo', 'giant otter', 'bighorn sheep', 'hippopotamus', 'california ground squirrel', 'european bee-eater', 'beech marten', 'leopard gecko', 'tailorbird', 'testudinidae', 'emperor penguin', 'northern pike', 'giant clam', 'stoat', 'horse', 'nutria', 'tree-kangaroo', 'giraffe', 'guinea baboon', 'ferret', 'bonytail chub', 'baya weaver', 'brook trout', 'pelican', 'mallard', 'roseate spoonbill', 'mountain weasel', 'pocket gophers', 'lybia edmondsoni', 'giant anteater', 'common raccoon dog', 'dewdrop spiders', 'armadillo girdled lizard', 'arctic fox', 'bison', 'swordfish', 'bald eagle', 'chimpanzee', 'asbolus verrucosus', 'sperm whale', 'abalone', 'golden jackal', 'hornet', 'zebra', 'orangutans', 'peregrine falcon', 'atlantic cod', 'burrowing owl', 'african wild dog', 'maned wolf', 'honey bee', 'naked mole-rat', 'echidnas', 'bowerbirds', 'rhinoceros', 'beaver', 'bombyx mori', 'common box turtle', 'hummingbird', 'domestic sheep', 'wolverine', 'raccoon', 'evergreen bagworm', 'pig', 'muskrat'}

D BIOSPARK DATASET PIPELINE EVALUATION: INCREASE IN ORGANISM DIVERSITY

In order to evaluate the effectiveness of diversification through our expansion strategies from iteratively constructed taxonomic trees, we investigated how organism diversity changes upon a series of mechanism generation.

D.1 Procedure

We generate mechanisms and corresponding species for five problems closely related to automobile design: 'managing impact', 'managing tension', 'managing compression', 'managing turbulence', 'modifying speed'. We index the species at the time of its appearance in the corresponding mechanism generation. Hence, the index corresponds to when a new mechanism was generated via our pipeline. At each generation index, we count the unique number of names that are generated up to that point, for each taxonomic rank, and average the numbers across the five problems.

D.2 The pattern of increasing organism diversity

Qualitatively we observe that the number of species generated are monotonically increasing (Fig. 8), albeit at a decreasing rate. The ratio between the number of unique species and the generation index is 2:1 at index=200, and approaches 4:1 near index=800. This suggests more mechanisms are generated and become concentrated on individual species (e.g., for grasshoppers, there might be several distinct mechanisms relevant to the problem of 'managing turbulence' such as their foldable wing structures, lightweight exoskeleton

[System Message]

You are an expert biologist who knows species and their taxonomic hierarchy very well. Follow the instructions to the letter.

- Return the scientific term for each taxonomic rank the species belongs to.
- Enclose keys and values using double quotes ("...") and format them into a Python dictionary.
- Use the taxonomic ranks as keys and corresponding scientific terms as their values.
- Do not add any other text.

[User Message]

What {"domain", "kingdom", "phylum", "class", "order", "family", "genus"} does "{organism}" belong to? Format your reply into a Python dictionary.

Figure 7: The prompt used to generate the taxonomy of each organism.

Model	DOMAIN	KINGDOM	PHYLUM	CLASS	ORDER	FAMILY	GENUS
GPT4	100%	100%	100%	100%	96.7%	94.4%	98.9%
	(90/90)	(90/90)	(90/90)	(90/90)	(87/90)	(85/90)	(89/90)
GPT3.5-turbo	100%	100%	100%	100%	95.6%	95.6%	93.3%
	(90/90)	(90/90)	(90/90)	(90/90)	(86/90)	(86/90)	(84/90)

Table 1: The accuracy of zero-shot taxonomy generation using only the organism name.

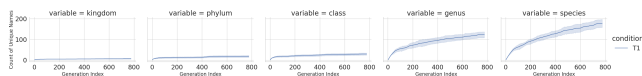


Figure 8: Organism diversity, measured by the number of unique names among the generation, increases monotonically as generation continues, while the number of mechanisms generated per organism also increases, as evidenced by the decreasing slope. This suggests mechanism concentration over each organism increases over time.

designs, or joints in their legs enabling repeated high jumps) on average as generation continues. In our future work, we will explore whether and how the mechanisms generated for the same species semantically differ from one another. In addition, we plan to examine how our generation approach compares to alternative curation (e.g., [14], which explored a data programming approach for mining concise biological problem-solving inspirations on Wikipedia) or generation approaches in terms of the efficiency of generating mechanisms across diverse organisms, by measuring the slope of organism diversity over generation index.

E ADDITIONAL CASE STUDY: ‘DESIGN ROBUST COLLISION RESISTANCE FOR RACING CARS’

For another designer working on improving collision resistance for racing cars, the core challenge might be the management of compression during a collision. When the designer clicks on the ‘Manage Collision’ problem as a query (Fig. 9, top right), BioSPARK once again displays biological mechanism clusters related to the problem. He finds “Interlocking barbules for structural integrity during flight” and “Pressure management in segmented skeletons,” interesting mechanisms employing different approaches to compression management (the former commonly found in feathers, the latter among isopods). He selects the ‘Combine’ tab to obtain design ideas that

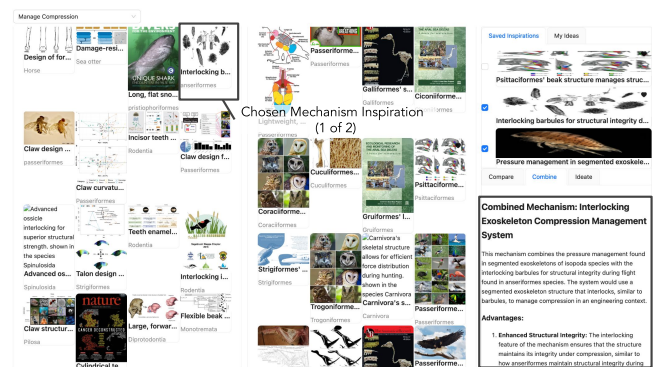


Figure 9: BioSPARK displays biological mechanisms for the query design challenge ‘Manage Compression’. User-chosen mechanisms are: “Interlocking barbules for structural integrity during flight” and “Pressure management in segmented skeletons.”

could benefit from both mechanisms. The result describes an approach that applies interlocking structures to exoskeleton segments, providing him with insight into a ‘sliding-lock’ mechanism for modular panels of the racing car body. This mechanism allows for extensive compression upon collision, thus increasing impact time (subsequently decreasing peak impact force) and more effectively dissipating force to non-critical body components. A video demonstrating user interaction can be also accessed at: <https://drive.google.com/file/d/1F2z1AX2aTMlpAzNX9kyePuMebFsB0-Wx/view?usp=sharing>.

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